

Module Theory Cheat-Sheet

For the most part, on this sheet I will assume rings to be commutative, with the one exception being when the matrix ring $M_2(k)$ is discussed.

Examples

- A $\mathbb{Z}$ -module is an abelian group.
- A $k[x]$ -module is a k -vector space V and a linear transformation $X: V \rightarrow V$.
- A $k[x, y]$ -module is a k -vector space V and *two* linear transformations $X, Y: V \rightarrow V$ such that $XY = YX$. (And so on for k adjoin several variables.)
- A module over $k[x]/(f)$ is a k -vector space V and a linear transformation $X: V \rightarrow V$ satisfying $f(X) = 0$. (That is, the minimal polynomial of X divides f .)
- The vector space k^2 is a left $M_2(k)$ -module, where a matrix A acts on a vector v by left matrix multiplication Av . In fact, every f.g. left $M_2(k)$ -module is a direct sum of k^2 with itself. This is related to the fact that $M_2(k)$ is a simple ring.

Constructions

- For any ring R , an ideal $I \subset R$ is an R -module, as is R/I .
- Given R -modules M, N , let $\text{Hom}_R(M, N)$ be the set of all R -module homomorphisms $f: M \rightarrow N$. Then $\text{Hom}_R(M, N)$ is itself an R -module, since if $f: M \rightarrow N$ is a homomorphism, so is $rf: M \rightarrow N$.
- If $R' \subset R$ is a subring, and M is an R -module, then M is moreover an R' -module. For example, a $\mathbb{C}[x]$ -module is also a $\mathbb{C}$ -vector space; a $\mathbb{C}$ -vector space is also an $\mathbb{R}$ -vector space of twice the dimension; a $\mathbb{Q}$ -vector space is also an abelian group (a $\mathbb{Z}$ -module). This way of forming modules is called **restriction of scalars**.
- More generally, if $f: R' \rightarrow R$ is any ring homomorphism, and M is an R -module, then we can view M as an R' -module via $r' \cdot m := f(r')m$. E.g. $k[t]$ can be a module over $k[x, y]/(y^2 - x^3)$ via the map $x \mapsto t^2, y \mapsto t^3$.

Modules over PIDs

Theorem. A finitely-generated module over a PID R is isomorphic to a module of the form

$$R^r \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m),$$

where each $a_i \mid a_{i+1}$. The elements a_i are called the **invariant factors** of the module. Alternatively, a f.g. module over a PID is isomorphic to a module of the form

$$R^r \oplus R/(p_1^{e_1}) \oplus R/(p_2^{e_2}) \oplus \cdots \oplus R/(p_n^{e_n}),$$

where each p_j is a prime element of R (which are not necessarily distinct). The p_j are called the **elementary divisors** of the module.